

Bruce W

AI Systems Architect · Agent Engineering · Platform Engineering

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PROFILE

AI systems architect with more than a decade of software engineering experience across Agent systems, evaluation infrastructure, recommendation platforms, enterprise financial software, and commerce systems. I turn fragmented engineering workflows into standardized, reusable platforms by identifying stable boundaries, encoding domain knowledge, and designing measurable quality and governance mechanisms.

SELECTED ARCHITECTURE IMPACT

- Architected an **Agent Evaluation Platform** supporting extensible offline and training-time evaluation, resource-aware scheduling, version governance, and multi-team traffic protection.
- Built an **Agent Engineering Knowledge System** with 7 reusable Skills; reduced benchmark integration from approximately 2 days to 10 minutes, while capturing 15 practical cases and 20+ engineering principles.
- Contributed to a **Configuration-Driven Recommendation Platform** that reduced model deployment from 4 hours to 30 minutes, improved measured inference QPS by 5.2×, and reduced estimated infrastructure cost by approximately 75% in the evaluated workload.

EXPERIENCE

Software Architect · AI Engineering — vivo Mobile Technology 2021–Present

- Lead architecture and delivery across Agent evaluation, language-model engineering, recommendation infrastructure, and AI engineering productivity.
- Evolved an internal evaluation framework into a shared platform with stable extension points, priority scheduling, reproducible releases, quotas, authentication, and controlled rollout.
- Designed human-in-the-loop Agent workflows that encode expert knowledge, automate validation, and retain human ownership of architecture and production approval.
- Contributed to configuration-driven feature and model serving used across multiple recommendation workflows, including heterogeneous CPU/GPU inference.

Software Architect — Shenzhen Fenqile Technology 2018–2020

- Designed commerce infrastructure for targeted promotions, coupon lifecycle management, redemption tracking, and real-time operational analytics.
- Integrated promotion services with existing commerce, payment, and data-synchronization systems.
- Contributed to patented work covering recommendation, configurable coupon presentation, and efficient data storage.

Senior Software Developer — IBM 2015–2017

- Delivered secure enterprise software for financial-services environments, covering transaction processing, risk workflows, and customer operations.
- Designed highly available integrations across APIs, databases, and third-party systems; mentored engineers and promoted maintainable practices.

Software Engineer — Chow Tai Fook Group 2011–2015

- Developed and maintained customer-facing retail web systems and contributed to a responsive website modernization.

SELECTED PROJECTS

Agent Evaluation Platform

Designed a platform spanning a stable evaluation core, strategy extensions, task and resource scheduling, version governance, and multi-team access controls. Introduced Agent-assisted self-service integration with automated review and human approval.

Agent Engineering Knowledge System

Converted expert workflows, decision trees, contracts, examples, common mistakes, and validation rules into reusable Agent capabilities. Added a human-confirmed knowledge flywheel that distills execution issues into reusable principles.

Configuration-Driven Recommendation Platform

Contributed to a shared Feature-as-a-Service and Model-as-a-Service architecture that aligned offline and online definitions and supported CPU/GPU serving behind stable interfaces.

TECHNICAL STRENGTHS

Architecture: Agent systems, AI evaluation, platform engineering, distributed systems, recommendation infrastructure, human-in-the-loop workflows

AI/ML Engineering: LLM workflows, model evaluation, feature platforms, model serving, TensorFlow, PyTorch, GPU inference

Software: Java, Python, JavaScript, C#, API design, extensible architecture, testing and validation

Infrastructure: Docker, Kubernetes, CI/CD, observability, resource scheduling, release and traffic governance

ADDITIONAL INFORMATION

- Four public patents spanning recommendation, commerce presentation, and data systems
- Two-time quarterly professional recognition
- Languages: Chinese and Cantonese (native/fluent); English (professional working proficiency)
- Bachelor's degree in Information Management and Information Systems

Public edition — selected personal details and precise education dates omitted.